1 35. On Maximal Domains of Integral Functions

The following investigations are closely connected with Hilbert's problem on relatively integral functions. They go back to an occasional question of E. Hecke, who needed a special case as an auxiliary lemma and handled it by direct calculation.

The question is this. By a maximal domain of integral functions in the variables $x_1, \ldots, x_n$ - polynomials, power series, or quotients of such - inside a given domain $\mathfrak{B}$, I mean a domain which can no longer be enlarged by adjoining integral denominators. Thus, whenever $a\, g(x)$ belongs to it, with $g(x) \in \mathfrak{B}$ and a a non-zero rational integer, then $g(x)$ itself must already belong to it.

Every integral domain $\mathfrak{A}$ in $\mathfrak{B}$ generates uniquely such a maximal domain $M(\mathfrak{A})$. It consists of all functions $g(x)$ in $\mathfrak{B}$ for which there is at least one non-zero integer a such that

$$ag(x) \in \mathfrak{A}.$$

The question is what can be said about the denominators a of this maximal domain when the original integral domain $\mathfrak{A}$ was finite, that is, when it consists of all integral polynomials in finitely many functions

$$f_1(x), \ldots, f_r(x)$$

from $\mathfrak{B}$.

For $\mathfrak{B}$ one may take all integral polynomials or power series in $x_1, \ldots, x_n$, or else those quotients which contain no denominators except those already occurring among the $f_i(x)$, and their powers. I show - under the additional hypothesis, in the case of power series or quotients of power series, that only algebraic relations occur among the $f_i(x)$ - that the integers a may always be chosen so that only finitely many different prime numbers occur in them. The powers of those primes, however, need not be bounded.

This last point is immediate. If $ag(x)$ belongs to $\mathfrak{A}$ but $a^\rho g(x)$ does not belong to $\mathfrak{A}$ for any fixed bound on ρ, then all powers of a may occur. Thus the reasonable boundedness question is different: can one give, in general, an infinite system of generators of $M(\mathfrak{A})$ in which the exponents of the primes in the corresponding denominators remain bounded? Since only finitely many primes are involved, this is equivalent, by known arguments, to the finiteness of $M(\mathfrak{A})$ itself. It is therefore Hilbert's problem on relatively integral functions in its integral form. The methods used here do not decide that question.

The proof that only finitely many different primes p occur in the denominators rests on the following idea. To every such prime there corresponds at least one relation modulo p among the functions $f_i(x)$ generating $\mathfrak{A}$, a relation with integral coefficients which is not identically satisfied in the variables x before reduction. In another language, let $\mathfrak{o}$ be the polynomial ring over the integers in indeterminates $u_1, \ldots, u_r$. The assignment

$$u_i \longmapsto f_i(x)$$

makes $\mathfrak{A}$ a homomorphic image of $\mathfrak{o}$; the homomorphism is defined by a prime ideal $\mathfrak{a}$ of $\mathfrak{o}$, so that

$$\mathfrak{A} \simeq \mathfrak{o}/\mathfrak{a}.$$

After reduction modulo p, $\mathfrak{B}_p$ is likewise a homomorphic image of the reduced polynomial ring $\mathfrak{o}_p$ except for the finitely many primes already appearing in the denominators of the $f_i(x)$. The new homomorphism is defined by a prime ideal $\mathfrak{c}_p$ of $\mathfrak{o}_p$ containing $\mathfrak{a}_p$.

A prime p occurs in the denominator of $M(\mathfrak{A})$ if and only if $\mathfrak{c}_p$ is a proper over-ideal of $\mathfrak{a}_p$. This can happen only if either the transcendence degree of $\mathfrak{B}_p$ drops in comparison with that of $\mathfrak{A}$, or if $\mathfrak{a}_p$ ceases to be a prime ideal. Both events occur only for finitely many primes. The first is controlled by functional determinants modulo p; the second by the fact that $\mathfrak{a}$ is an absolute prime ideal and can lose primality only modulo finitely many primes.

All arguments remain valid, with minor modifications, if the rational integers are replaced by the integers of a finite algebraic number field and the rational primes by the prime ideals of that field.

1.1 §1. Functional determinants over coefficient fields of characteristic p

Let

$$f_1(x_1, \ldots, x_n), \ldots, f_r(x_1, \ldots, x_n)$$

be rational functions, or more generally formally defined power series or quotients of such, with coefficients in a field P of characteristic p. Assume that the derivatives $\partial f_i/\partial x_j$ are formally defined and satisfy the usual rules of calculation. Let Ω be an algebraically closed extension field of P.

Then the following statement holds as in characteristic zero. If there is an algebraic dependence among the $f_i(x)$ with coefficients in Ω, then the rank of the functional matrix

$$\left(\frac{\partial f_i}{\partial x_j}\right)$$

is smaller than the number of algebraically independent functions under consideration, unless the dependence is purely inseparable. More precisely, if the functions are algebraically dependent and $G(u_1, \ldots, u_r)$ is a non-zero polynomial relation of least degree, then

$$G(f_1, \ldots, f_r) = 0.$$

Differentiation gives

$$\sum_i \frac{\partial G}{\partial u_i}(f)\frac{\partial f_i}{\partial x_j} = 0$$

for every j. If the functional matrix had full rank, all partial derivatives $\partial G/\partial u_i$ would vanish. In characteristic p, this means that G is a polynomial in the p-th powers,

$$G(u) = H(u_1^p, \ldots, u_r^p).$$

Since Ω is algebraically closed, this contradicts the choice of G as an irreducible polynomial of least degree. Thus the rank must drop.

The consequence needed below is the following. Let

$$F_1(x), \ldots, F_t(x)$$

be integral functions - polynomials, power series, or quotients of such - whose functional matrix has rank exactly t. Choose a prime p which occurs neither in the denominators of the $F_i(x)$ nor in all t-rowed determinants of the functional matrix. Then the reduced functions

$$f_1(x), \ldots, f_t(x)$$

modulo p remain algebraically independent.

Indeed, their functional matrix is obtained by reducing the entries of the original functional matrix modulo p. By the choice of p, the rank remains t. Algebraic dependence would force the rank to be smaller, by the preceding argument.

1.2 §2. Formulation of the theorem

Let $\mathfrak{A}$ be an integral domain of integral functions in $x_1, \ldots, x_n$: polynomials, power series, or quotients of such, with rational integral coefficients. In the case of power series one may understand formal power series or convergent power series; only the fact that the functions form an integral domain is used.

Let $\mathfrak{Q}$ be the quotient field of $\mathfrak{A}$, and let $\mathfrak{B}$ be an integral domain between $\mathfrak{A}$ and $\mathfrak{Q}$. The domain $\mathfrak{M}$ is called maximal in $\mathfrak{B}$ if the following condition holds: whenever $a \neq 0$ is an integer, $g(x) \in \mathfrak{B}$, and $ag(x) \in \mathfrak{M}$, then $g(x) \in \mathfrak{M}$.

The maximal domain generated by $\mathfrak{A}$ in $\mathfrak{B}$ exists uniquely. It is the intersection of all maximal domains in $\mathfrak{B}$ containing $\mathfrak{A}$, and it consists exactly of those $g(x) \in \mathfrak{B}$ for which $ag(x) \in \mathfrak{A}$ for at least one non-zero integer a.

Assume now that $\mathfrak{A}$ is finite and has a unit element. Let

$$f_1(x), \ldots, f_r(x)$$

be an integral basis, so that $\mathfrak{A}$ consists of all integral polynomials in these functions. Let $\mathfrak{B}$ consist of the allowed polynomials, power series, or quotients, with no denominators except those appearing among the finitely many $f_i(x)$.

In the case of power series or quotients one adds the following condition. If t is the rank of the functional matrix of the $f_i(x)$, and if, after renumbering, $f_1, \ldots, f_t$ already have a functional matrix of rank t, then the quotient field $\mathfrak{Q}$ is to be a finite algebraic extension of

$$\mathbb{Q}(f_1, \ldots, f_t).$$

Equivalently, the remaining $f_{t+1}, \ldots, f_r$ are algebraic over the field generated by $f_1, \ldots, f_t$.

Let $\mathfrak{o} = \mathbb{Z}[u_1, \ldots, u_r]$. Then $\mathfrak{A}$ is a homomorphic image of $\mathfrak{o}$ under $u_i \mapsto f_i(x)$, hence

$$\mathfrak{A} \simeq \mathfrak{o}/\mathfrak{a}$$

for a prime ideal $\mathfrak{a}$. The theorem says that, except for finitely many rational primes p, reduction modulo p gives no new denominator obstruction: if a is not divisible by any of those exceptional primes and $ag(x)$ lies in $\mathfrak{A}$, then $g(x)$ already lies in $\mathfrak{A}$.

Equivalently, only finitely many primes can occur in the denominators required to pass from $\mathfrak{A}$ to its maximal domain.

1.3 §3. Proof of the theorem

The first auxiliary assertion is that the prime ideal $\mathfrak{a}$ which defines the homomorphism from $\mathfrak{o}$ to $\mathfrak{A}$ is absolutely prime, under the stated hypotheses. In other words, after extending the coefficient field to an algebraic closure, the corresponding ideal remains prime. This is a standard consequence of the irreducibility of the system $f_1, \ldots, f_t$ and of the algebraic dependence of the remaining functions over it.

Therefore $\mathfrak{a}$ can cease to be prime after reduction modulo p for at most finitely many primes. This is one part of the exceptional set. A second finite exceptional set is formed by the primes in the denominators of the $f_i(x)$. A third is formed by the primes at which all suitable t-rowed functional determinants vanish modulo p. Outside these finitely many primes, three facts hold simultaneously:

1. the reduction of the functions $f_i(x)$ is defined;
2. the reduced ideal $\mathfrak{a}_p$ is still prime;
3. the reduced functions $f_1, \ldots, f_t$ remain algebraically independent.

Choose p outside the exceptional set. The homomorphism from the reduced polynomial ring $\mathfrak{o}_p$ to $\mathfrak{B}_p$ is defined by a prime ideal $\mathfrak{c}_p$ containing $\mathfrak{a}_p$. It remains to show that in fact

$$\mathfrak{c}_p = \mathfrak{a}_p.$$

For this it is enough to compare transcendence degrees. The residue field of $\mathfrak{c}_p$ is the quotient field of the reduced image domain, so its transcendence degree is at least t. The residue field of $\mathfrak{a}_p$ also has transcendence degree exactly t, because the classes of $u_1, \ldots, u_t$ remain algebraically independent and the remaining classes are algebraic over them.

To see the latter point explicitly, use the algebraic relations in characteristic zero. The ideal $\mathfrak{a}$ contains primitive integral polynomials

$$G_{t+1}(u_1, \ldots, u_t, u_{t+1}), \ldots, G_r(u_1, \ldots, u_t, u_r)$$

expressing the algebraic dependence of the remaining generators. For all but finitely many primes these polynomials do not vanish after reduction and still involve the corresponding variable u_j genuinely. Hence the classes of $u_{t+1}, \ldots, u_r$ are algebraic over the classes of $u_1, \ldots, u_t$ modulo $\mathfrak{a}_p$.

Thus $\mathfrak{a}_p$ and $\mathfrak{c}_p$ are prime ideals of the same transcendence degree, and $\mathfrak{c}_p$ contains $\mathfrak{a}_p$. A prime ideal has no proper prime over-ideal of the same transcendence degree. Therefore $\mathfrak{c}_p = \mathfrak{a}_p$.

This proves that no such prime p can occur in a denominator. The theorem follows.

1.4 §4. Algebraic integer coefficient domains

The same reasoning applies if the rational integers are replaced by the ring of integers of a finite algebraic number field K, and rational primes by prime ideals of K.

The determinant argument and the formulation of the theorem are unchanged. In the proof, however, one must add to the exceptional prime ideals those which divide the finitely many algebraic integers occurring as contents of the polynomials expressing the algebraic dependences; these polynomials can no longer all be assumed primitive over $\mathbb{Z}$ in the same naive sense.

The final formulation is as follows. Let a be an integer of K not divisible by any of the finitely many exceptional prime ideals. If $ag(x)$ lies in the original domain $\mathfrak{A}$ and $g(x) \in \mathfrak{B}$, then $g(x)$ lies in the maximal domain generated by $\mathfrak{A}$ without needing any of the non-exceptional prime ideals as denominators.

Equivalently, after localizing away from the exceptional prime ideals, the same argument as above shows that the relevant reduced homomorphisms have the same prime ideal kernel. If

$$(a) = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_s^{e_s}$$

is the prime-ideal factorization and none of the $\mathfrak{p}_i$ is exceptional, the localization makes these prime ideals principal and leaves the ideal decomposition unchanged. The preceding proof then applies word for word.

If the exceptional prime ideals are generated in the localized ring by elements

$$\lambda_1, \ldots, \lambda_q,$$

then every needed denominator may be chosen as a product of powers of these finitely many λ_i. Returning from the localized ring to the original ring of integers gives the asserted finite-prime statement.